Double-Angle Formulas and Trigonometric Equations

Section 1: Double-Angle Formulas

1 State the double-angle formula for $\sin(2\theta)$.

a. $2\sin\theta\cos\theta$

b. $\cos^2\theta - \sin^2\theta$

c. $2\sin\theta$

d. $\sin^2\theta$

2 Which of the following is NOT an equivalent form of $\cos(2\theta)$?

a. $1 - 2\cos^2\theta$

b. $\cos^2\theta - \sin^2\theta$

c. $2\cos^2\theta - 1$

d. $1 - 2\sin^2\theta$

3 If $\cos\theta = 4/5$ with θ in Quadrant IV, find $\sin(2\theta)$.

a. $-24/25$

b. $24/25$

c. $-7/25$

d. $7/25$

Section 2: Half-Angle and Power-Reducing Formulas

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4 State the power-reducing formula for $\sin^2 x$.

a. $\frac{1-\cos(2x)}{2}$

b. $\frac{1+\cos(2x)}{2}$

c. $\frac{1-\sin(2x)}{2}$

d. $1-\cos^2 x$

5 Find the exact value of $\cos(22.5^\circ)$ using the half-angle formula $\cos(\theta/2)=\sqrt{\frac{1+\cos\theta}{2}}$.

a. $\frac{\sqrt{2+\sqrt{2}}}{2}$

b. $\frac{\sqrt{2-\sqrt{2}}}{2}$

c. $\frac{\sqrt{2}+1}{2}$

d. $\frac{\sqrt{3}+1}{2}$

6 Rewrite $\cos^4 x$ in terms of first powers of cosines of multiple angles.

a. $\frac{3}{8}+\frac{1}{2}\cos(2x)+\frac{1}{8}\cos(4x)$

b. $\frac{1}{4}(1+\cos(2x))^2$

c. $\frac{1}{2}+\frac{1}{2}\cos(4x)$

d. $\frac{3}{8}-\frac{1}{2}\cos(2x)$

Section 3: Solving Linear Trigonometric Equations

7 Solve $2\sin x-1=0$ on the interval $[0,2\pi)$.

a. $\frac{\pi}{6}$ and $\frac{5\pi}{6}$

b. $\frac{\pi}{3}$ and $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$ and $\frac{11\pi}{6}$

d. $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

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8 Solve $\sqrt{3}\tan x + 1 = 0$ on $[0, 2\pi)$.

a. $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

b. $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

c. $\frac{\pi}{6}$ and $\frac{7\pi}{6}$

d. $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

9 Find all solutions to $\cos(2x) = 0$ on $[0, \pi)$.

a. $\frac{\pi}{4}$ and $\frac{3\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{4}$ only

d. $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

Section 4: Quadratic-Type and Multiple-Angle Equations

10 Solve $2\sin^2 x - \sin x - 1 = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

c. $\frac{\pi}{2}, \frac{5\pi}{6}$

d. $\frac{\pi}{3}, \frac{5\pi}{3}$

11 Solve $\sin(2x) = \cos x$ on $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{2}, \frac{3\pi}{2}$

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12 Solve $\cos(2x) + \cos x = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

b. $\frac{\pi}{6}, \pi, \frac{11\pi}{6}$

c. $\frac{\pi}{2}, \frac{3\pi}{2}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

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